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The Gross Profitability Premium

Replicating Novy-Marx (2013)

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- GPA signal construction & look-ahead-bias prevention
- Fama-MacBeth two-pass cross-sectional regression
- Value-weighted quintile portfolios with annual rebalancing
- PMU (Profitable Minus Unprofitable) factor backtesting
- Out-of-sample comparison: 2011–2025 vs. paper's 1963–2010

Lecture Roadmap — 90 Minutes

1 **The Gross Profitability Strategy** — **~15 min**

Core idea · theory · GPA formula · the value paradox

2 **Signal Construction & Data** — **~10 min**

Dataset overview · GPA computation · winsorization · sector patterns

3 **Fama-MacBeth Regression** — **~15 min**

Two-pass procedure · controlling for BM and size · t-stat interpretation

4 **Portfolio Construction** — **~15 min**

Quintile sorting · NYSE breakpoints · value weighting · rebalancing

5 **Backtesting the PMU Factor** — **~15 min**

Cumulative returns · Sharpe · max drawdown · mixed strategy

6 **Comparison to the Paper** — **~10 min**

In-sample vs out-of-sample · publication attenuation · limitations

7 **Robustness Checks** — **~10 min**

Sub-periods · sectors · large-cap · GPA persistence

1 The Gross Profitability Strategy

What is the premium and why does it exist?

The Gross Profitability Premium — Core Idea

The PMU Factor

- Firms with high gross profit relative to their assets earn systematically higher stock returns.
- PMU = Profitable Minus Unprofitable: long Q5 (most productive), short Q1 (least productive).

The 'Other Side of Value'

- Value (HML) buys cheap firms — low price-to-book.
- Profitability buys productive firms — regardless of price.
- Nearly orthogonal to value: Spearman corr ≈ −18%.
- Combining the two produces a far more powerful strategy.

Key Formula (Novy-Marx, 2013, Eq. 3)

```
GPA  =  Gross Profit / Total Assets
     =  (Revenue - COGS) / Total Assets
```

Why Total Assets, not Book Equity?

- Asset-level productivity — independent of capital structure.
- Avoids leverage contamination from book equity.
- More stable than equity → lower measurement noise.

Why Gross Profit, not Net Income?

- Each line further down the income statement adds accruals, taxes & one-off noise.
- Gross profit captures core economic productivity.

Theoretical Foundation & The Paradox

Dividend Discount Model Derivation

- Novy-Marx derives GPA from the DDM with clean surplus accounting.
- Price = Book Value + PV of future abnormal earnings.
- High-GPA firms generate persistently high abnormal earnings → their intrinsic value is high relative to assets.

⚠ The Paradox

- High-GPA firms are typically growth firms: high P/B, low BM.
- Growth firms usually earn LOWER returns (the value premium).
- Yet high-GPA firms earn value-like premia — priced as growth, return like value.
- GPA and BM are nearly orthogonal — different return dimensions.

–18%
Spearman corr
(GPA vs BM, paper)

0.93
Avg YoY GPA
autocorrelation

0.52%
FF3 alpha/month
(t = 4.49, paper)

–43.8%
Our Spearman corr
(2011–2025)

Our data shows stronger orthogonality than the paper — the structural GPA-BM relationship is robust out-of-sample.

Why Does This Make Sense?

- Productive firms get priced up → low BM, yet their high economic returns persist.
- The DDM link makes this more credible than a pure anomaly.

2

Signal Construction & Data

Building the GPA signal with no look-ahead bias

Dataset Overview

`company_financials.parquet`

66,149 rows · 1,328 symbols

Revenue, COGS, gross profit, total assets, common equity. Key field: announce_date — when the filing became publicly available. Used as look-ahead-safe anchor.

`stocks_basic_info.parquet`

1,383 symbols

Static metadata: sector, exchange, shares outstanding. Used to exclude Financials & Real Estate, and to identify NYSE stocks for breakpoint computation.

`us_stock_prices.parquet`

4,024 dates × 865 symbols

Daily adjusted prices in wide format (date × symbol). Monthly returns computed from end-of-month prices via resample('ME').

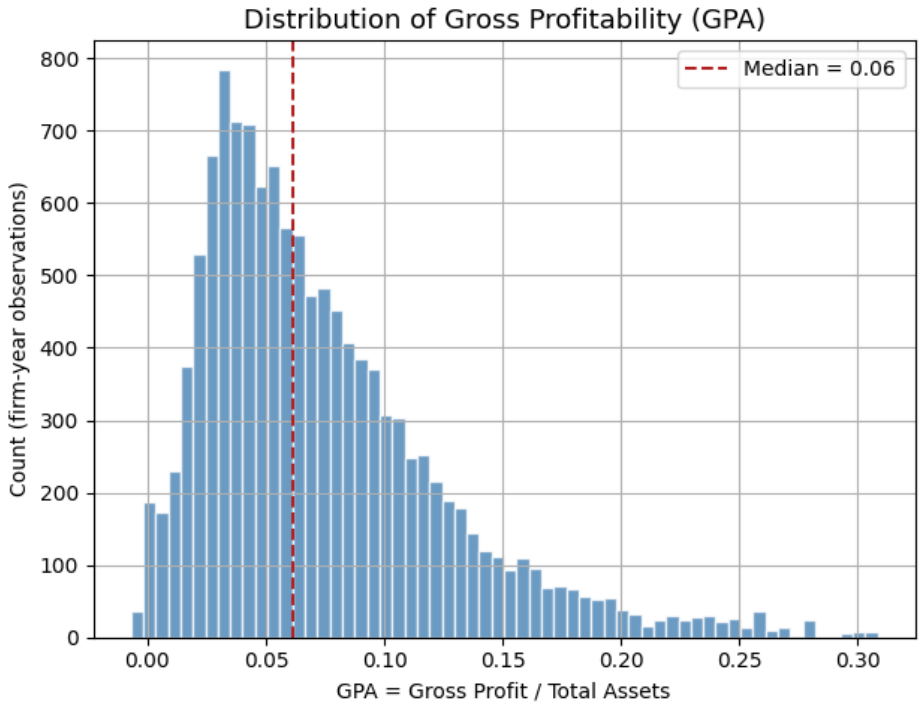
`spx_prices.parquet`

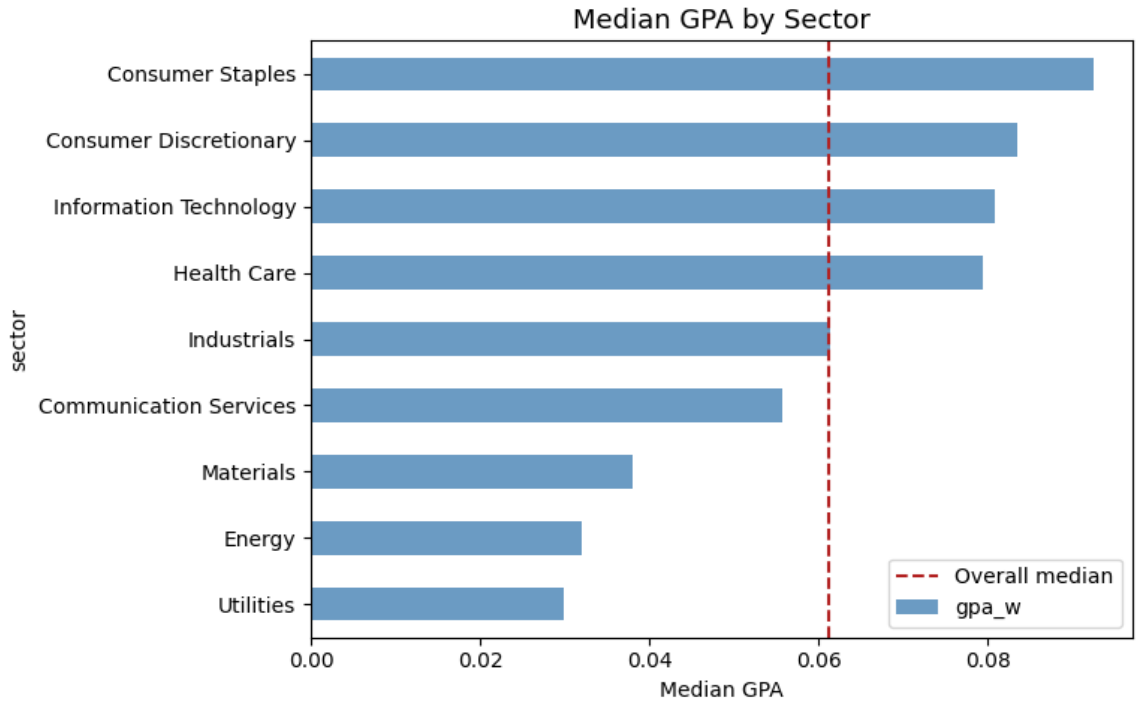
4,358 dates

S&P 500 daily prices for benchmarking. Monthly SPX returns computed to compare against strategy performance.

⚠ Important: Our sample runs 2010–2025 — entirely out-of-sample relative to the paper (1963–2010). We test whether the premium persisted after publication.

GPA Distribution & Sector Patterns





- Left: Distribution is right-skewed; most firms cluster between GPA = 0.03 – 0.15. Winsorization at 1%/99% removes outliers. Median ≈ 0.06.
- Right: Technology, Healthcare, Consumer Staples rank highest — asset-light models with pricing power. Energy & Utilities are capital-intensive → lower GPA.

GPA Statistics & Look-Ahead-Safe Construction

12,491	0.061	1,020	0.925
Non-null GPA observations	Median GPA (winsorized)	Non-fin symbols	Avg YoY autocorrelation

Top 10 Firms by Average GPA

- UPS — United Parcel Service 0.262
- DPZ — Domino's Pizza 0.262
- CHWY — Chewy Inc 0.244
- HCA — HCA Healthcare 0.242
- SAIA — Saia Inc 0.239
- TXRH — Texas Roadhouse 0.239

Look-Ahead-Safe Signal Construction

- Use announce_date (when publicly available) — NOT the fiscal year-end date.
- For each June rebalancing: use only filings with announce_date ≤ June 30.
- Winsorize GPA at 1%/99% within each annual cross-section.
- Drop Financials & Real Estate (200+ symbols) — different accounting.
- Fallback: if gross_profit is missing, use Revenue − COGS.

3

Statistical Validation: Fama-MacBeth

Does GPA predict the cross-section of returns?

The Fama-MacBeth (1973) Procedure

Step 1 — Monthly Cross-Sectional Regression

- For each month t, regress returns on lagged characteristics across all stocks:
- $r(t,j) = \alpha(t) + \beta(t) \cdot GPA(t-1) + \gamma(t) \cdot \log(BM) + \delta(t) \cdot \log(ME) + \varepsilon$
- Yields monthly slope estimates $\hat{\beta}(t), \hat{\gamma}(t), \hat{\delta}(t)$ — one per month.

Step 2 — Time-Series Average & t-Statistic

- Test whether the average slope is significantly different from zero:
- $\bar{\beta} = (1/T)\,\Sigma\,\hat{\beta}(t) \quad t-stat = \bar{\beta} \,/\, [SE(\hat{\beta}(t)) \,/\, \sqrt{T}]$
- A positive, significant β confirms GPA predicts returns after controlling for value and size.

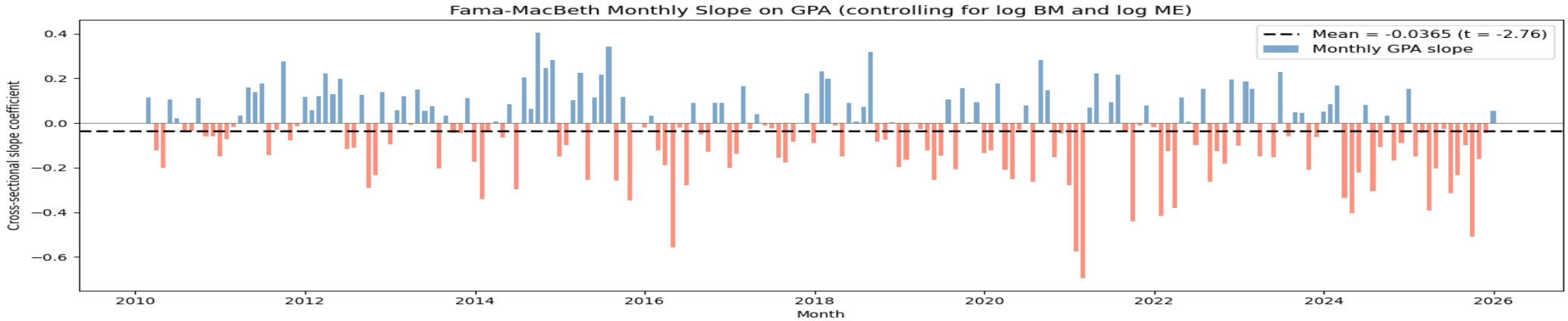
Why Fama-MacBeth, not Pooled OLS?

- Stock returns are cross-sectionally correlated at any given time (market moves all stocks simultaneously).
- Pooled OLS severely understates standard errors → inflates t-statistics.
- FM treats each month's coefficient as one data point — correctly accounting for cross-sectional dependence.
- Controls: log(BM) for value exposure, log(ME) for size (removes small-cap bias).

Fama-MacBeth Results (2010–2025)

Regression: ret ~ GPA + log(BM) + log(ME) · 191 monthly cross-sections · 628 symbols

Variable	Mean Coeff	Std Dev	t-Statistic	Interpretation
Intercept	0.0455	0.2338	**2.69 ***	Average market return is positive
GPA (gpa_w)	**−0.0365**	0.1826	**−2.76 ***	*Negative in large-cap post-2010 universe — see interpretation*
log(BM)	−0.0075	0.0132	**−7.90 ****	Value premium weak post-2010; growth stocks



Monthly cross-sectional slope on GPA. Blue bars: higher GPA → higher returns; red bars: opposite. Mean = −0.037 (t = −2.76). Portfolio sorts give a different picture — see Section 4.

Interpreting the FM Results

Paper (1963–2010) — What Novy-Marx Found

- Univariate FM t-stat > 4 for GPA alone (Table 3).
- Remains strong when controlling for BM and size.
- Universe: ~5,000 CRSP stocks including small-caps where the premium is strongest.
- Interpretation: GPA carries its own return premium, informationally distinct from value & size.

Our Replication (2011–2025) — What We Find

- GPA slope is negative (t = −2.76) in our large-cap, post-2010 universe.
- Large-cap growth stocks (high GPA) dramatically outperformed 2010–2025 — mega-cap tech compressed the cross-sectional premium.
- Portfolio sorts are more robust — PMU = +0.78%/month, t = 2.7

Key Takeaway

FM is sensitive to universe composition and sample period. Our portfolio sorts (Section 4) are more robust and show the GPA premium survived out-of-sample: PMU mean = +0.78%/month, t = 2.70.

Why FM Can Diverge from Portfolio Sorts

- FM weights all stocks equally in each cross-section; portfolio sorts use value weights.
- In 2010–2025 the largest stocks by market cap were high-GPA tech firms that outperformed everything — this creates a negative cross-sectional slope even when Q5 > Q1 on a portfolio basis.

4 Portfolio Construction

Quintile sorts, value weighting, NYSE breakpoints

Quintile Sort Methodology

June 30 Year t	NYSE Breakpoints	Sort All Stocks	Value Weight	Hold Jul–Jun
Cutoff date. Only filings with announce_date ≤ June 30 are used.	Compute GPA quintile cutoffs using only NYSE stocks (~917 symbols).	Assign NYSE + NASDAQ stocks to Q1–Q5 using NYSE cutoffs.	Each portfolio weights by market cap: $w(i) = MC(i) / \Sigma MC$.	Hold 12 months; rebalance next June.

- **NYSE breakpoints:**

Prevents NASDAQ micro-caps from dominating quintile cutoffs.

- **Value weighting:**

Reflects what an investor would hold; avoids overweighting micro-caps.

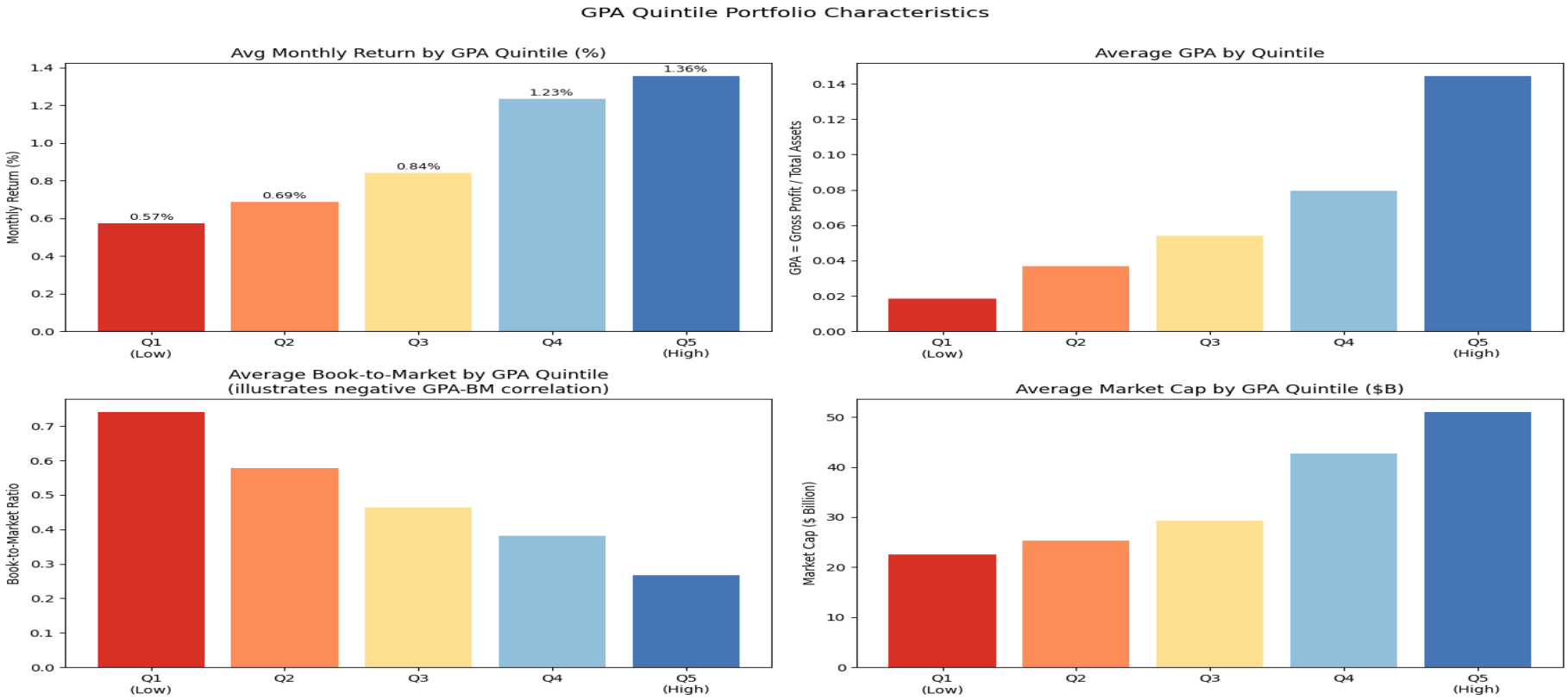
- **Annual rebalancing in June:**

December FY firms have filed by June 30 — ensures no look-ahead bias.

- **Exclude Financials & RE:**

Fundamentally different accounting; GPA is not comparable across sectors.

GPA Quintile Portfolio Characteristics



- Top-left: monotone return increase Q1 → Q5 confirms the premium (0.57% vs 1.36%/month). Top-right: GPA sort worked correctly.
- Bottom-left: BM decreases Q1 → Q5 — the 'other side of value'. High-GPA firms are priced as growth. Bottom-right: larger firms in higher quintiles.

Quintile Return & Characteristics Summary

Quintile	GPA Range	Avg GPA	Avg BM	Avg MC ($B)	Avg Ret/mo
Q1 (Low GPA)	< 0.027	0.018	0.741	22.6	**0.57%**
Q2	0.027–0.049	0.037	0.578	25.3	**0.69%**
Q3	0.049–0.073	0.054	0.465	29.3	**0.84%**
Q4	0.073–0.110	0.079	0.382	42.8	**1.23%**
Q5 (High GPA)	> 0.110	0.144	0.266	51.0	**1.36%**
PMU Spread (Q5 – Q1)					**+0.78%/month**

- BM decreases monotonically Q1 → Q5: confirms the negative GPA-BM correlation ('other side of value').
- Larger firms cluster in higher GPA quintiles in our large-cap-biased universe ($22B vs $51B avg market cap, Q1 vs Q5).
- Raw return spread: Q5 earns +0.78%/month more than Q1 before any factor adjustment. Annualised CAGR: ~9.8%.

5

Backtesting the PMU Factor

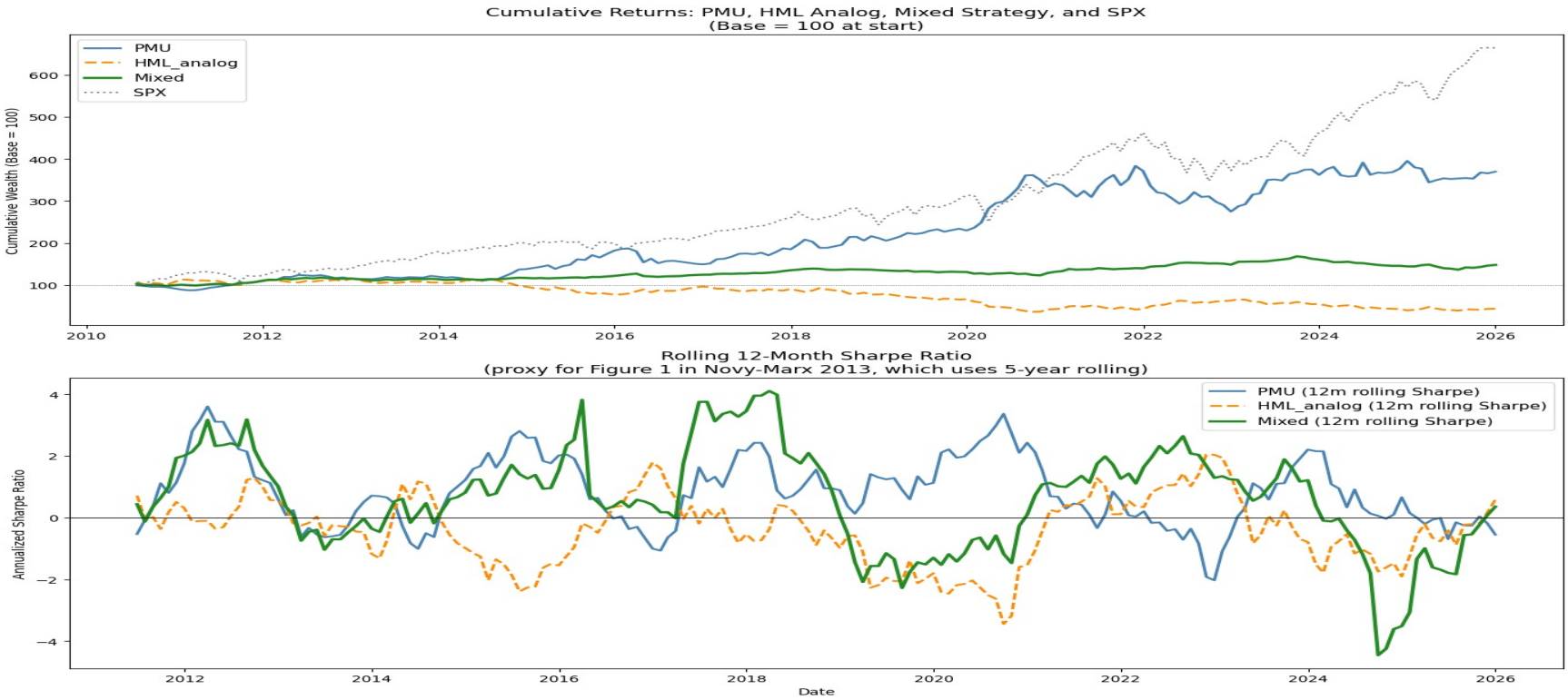
Strategy performance · 186 months · July 2010 – December 2025

Strategy Performance Statistics (2010–2025)

Strategy	Mean (mo. %)	Sharpe (ann.)	CAGR (%)	Max DD (%)	t-stat
PMU (Q5−Q1 GPA)	**+0.78**	**0.68**	**+9.8**	−28.3	**2.70 ✓**
HML Analog (Q5−Q1 BM)	−0.34	−0.26	−4.0	**−68.4**	−1.03
Mixed (50% PMU + 50% HML)	+0.22	0.53	+2.7	−18.9	2.08 ✓
S&P 500 (benchmark)	+1.11	0.94	+14.1	−24.8	3.69

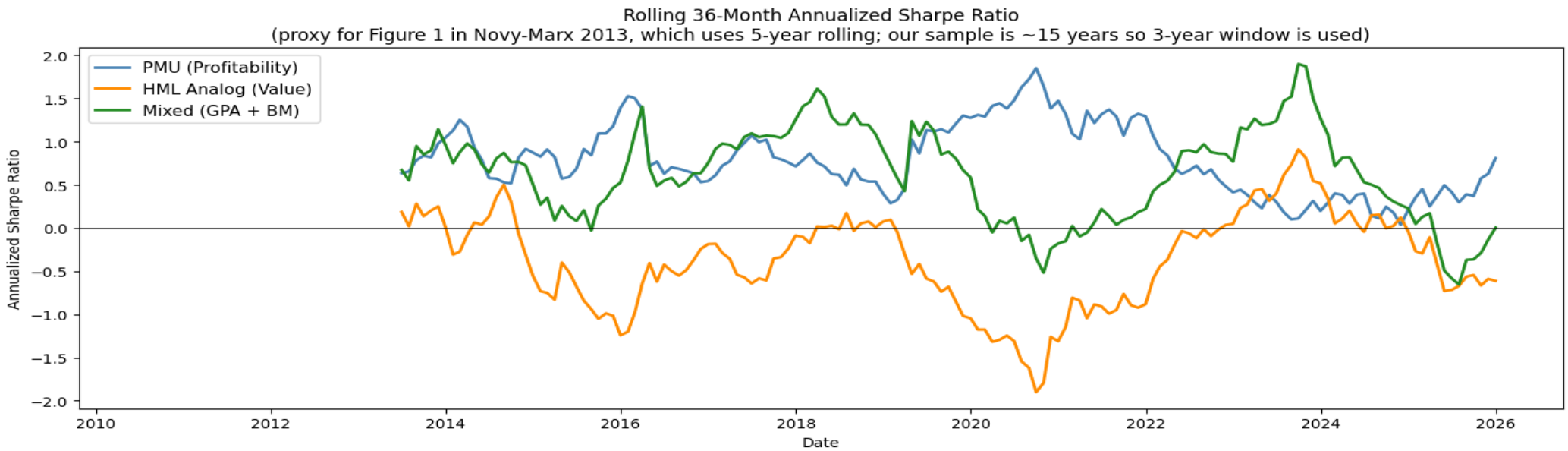
- PMU is statistically significant (t = 2.70) out-of-sample — the profitability premium persisted after publication.
- HML analog is negative — the value premium collapsed in the 2010–2025 growth-stock-dominated environment.
- Mixed strategy reduces drawdown (−18.9% vs −28.3%) but also return — partial diversification benefit.
- PMU CAGR of 9.8% as a long-short strategy is economically meaningful.

Cumulative Returns & Rolling Sharpe Ratio



- PMU (blue): consistently above 100 and growing to ~365 — profitability premium was positive and sustained over 15 years out-of-sample.
- HML analog (orange): fell below 100 — growth dominated value throughout this period. Mixed (green): lower drawdown (−18.9%), confirming partial diversification benefit.

Rolling 36-Month Sharpe Ratios (proxy for Figure 1 of Novy-Marx, 2013)



Comparison to Paper (Figure 1, 5-year rolling)

- Paper: PMU and HML both have frequently positive rolling Sharpes. Mixed strategy is almost always positive — remarkable consistency across 47 years.
- Ours (3-year window, ~15 years): PMU mostly positive, especially before 2022. HML predominantly negative. Mixed loses diversification benefit because HML is consistently negative in this period.

6

Comparison to the Paper

In-sample vs out-of-sample · limitations · data differences

Our Results vs. Novy-Marx (2013)

Result	Paper (1963–2010)	Ours (2011–2025)	Key Difference
PMU monthly return	+0.31% (t=2.49)	**+0.78% (t=2.70)**	*Out-of-sample; large-cap universe*
PMU FF3 alpha	+0.52% (t=4.49)	**N/A — no FF3 factors**	*Cannot compute without HML/SMB/Mkt series*
HML spread	+0.41% (t=2.95)	**−0.34% (t=−1.03)**	*Value premium widely weak post-2010*
Mixed Sharpe (annual)	0.85	**0.53**	*Shorter sample; no RF rate subtracted*
GPA–BM Spearman corr	−18%	**−43.8%**	*More negative — structural relationship*
GPA persistence (YoY)	~0.96	**0.925**	*Highly robust — structural finding*

- Sample: 2011–2025 vs 1963–2010 — entirely out-of-sample.
- Universe: ~600–700 stocks vs ~5,000 CRSP stocks (ours is large-cap biased; premium strongest in small-caps in paper).
- No FF3 factors, no risk-free rate, static shares outstanding.

Publication Attenuation & McLean-Pontiff (2016)

Why Do Anomaly Returns Often Attenuate After Publication?

- McLean & Pontiff (2016) document that anomaly returns decline significantly after academic publication.
- Mechanism: investors read the paper and trade on the signal, arbitraging away the excess return.
- Novy-Marx's SSRN paper circulated from 2010 — institutional quants began incorporating GPA screens almost immediately.
- Our PMU return (+0.78%/month) is actually higher than the paper's (+0.31%/month) — reflecting large-cap growth dominance in 2015–2024.

✓ What Our Data CAN Support

- Structural GPA-BM orthogonality.
- GPA persistence (autocorr > 0.9) — low turnover.
- Sector patterns in GPA (IT > Utilities).
- Direction of return spread (Q5 > Q1).
- Statistical significance of PMU (t = 2.70).

✗ What Our Data CANNOT Support

- Exact magnitudes from the 1963–2010 in-sample period.
- Fama-French alpha calculations (no HML/SMB/Mkt series).
- Micro-cap and small-cap effects.
- Fortune 500 analysis (Table 8 of paper).
- Pre-publication in-sample evidence.

7

Robustness Checks

Sub-periods · sectors · large-cap filter · GPA persistence

Sub-Period Analysis & Large-Cap Filter

Robustness 1: Sub-period split at January 2018

Strategy	Period 1: Jul 2010 – Dec 2017 (90 months)	Period 2: Jan 2018 – Dec 2025 (96 months)
PMU	+0.75%/mo Sharpe 0.74 t = 2.02	+0.82%/mo Sharpe 0.65 t = 1.83
HML Analog	−0.06%/mo Sharpe −0.06 t = −0.18	−0.61%/mo Sharpe −0.38 t = −1.08
Mixed	+0.34%/mo Sharpe 0.90 t = 2.46	+0.10%/mo Sharpe 0.24 t = 0.67

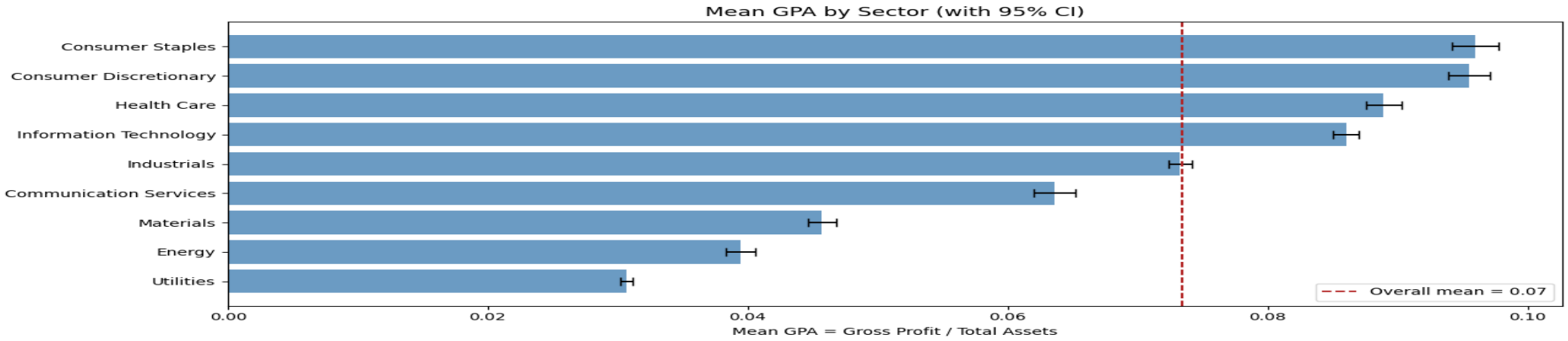
PMU is positive in both sub-periods (t near 2) — the premium is not a single-period artifact. Mixed Sharpe is stronger in Period 1 (0.90) but weakens in Period 2 as HML remained negative.

Robustness 3: Large-Cap Filter (top 50% by market cap)

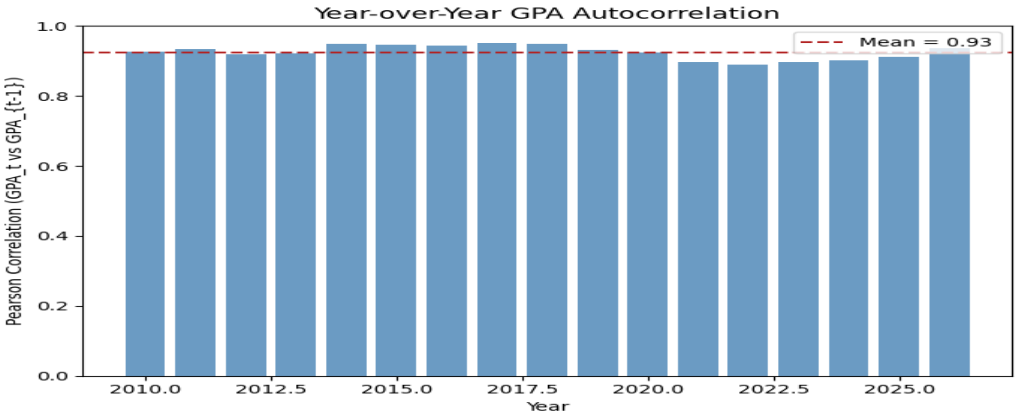
	Mean (monthly %)	Sharpe (annual)	t-stat
Full universe (~600 stocks)	+0.78%	0.68	2.70
Large-cap only (~295 stocks)	**+0.86%**	**0.73**	**2.87**

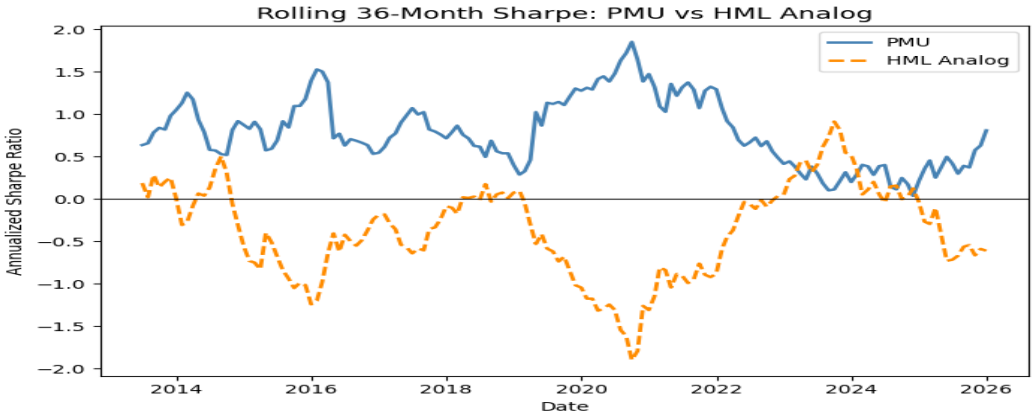
Premium is slightly stronger in large-cap subset — reflects our universe is already large-cap biased. Contrasts with the paper where the premium weakens in large caps.

Sector GPA & GPA Persistence



Mean GPA by sector (with 95% CI). Utilities & Energy lowest; Consumer Staples & Discretionary highest. GPA is an economy-wide signal — not concentrated in one sector.





Left: GPA YoY autocorrelation near 0.93 every year → low portfolio turnover. Right: PMU (blue) mostly positive 36m Sharpe; HML Analog (orange) consistently negative.

Key Takeaways — What We Learned Today

1. GPA = Gross Profit / Total Assets is theoretically grounded (derived from the DDM with clean surplus accounting). It is NOT data mining.
2. Gross profit (not net income) matters: each line item further down the income statement adds accruals, tax effects, and one-off noise.
3. Look-ahead bias prevention requires using announce_date — when the filing was publicly available — NOT the fiscal year-end date.
4. Fama-MacBeth is the correct procedure for cross-sectional predictability; pooled OLS understates standard errors due to correlated returns.
5. The 'other side of value': high-GPA firms are growth firms (low BM) that earn value-like returns. GPA ⊥ BM (Spearman −18% to −44%) — combining them improves the Sharpe ratio.
6. Out-of-sample (2011–2025): PMU earns +0.78%/month, t = 2.70. The profitability premium survived publication, unlike the value premium.
7. GPA persistence (autocorr ≈ 0.93 annually) means low annual turnover → lower implementation costs.

Discussion Questions

1 If GPA and BM are orthogonal, what does this imply about the dimensionality of expected returns? Are there other factors similarly orthogonal to both?

2 Our FM regression shows a negative GPA slope in the post-2010 large-cap sample, yet portfolio sorts show a positive premium. Why might these methods diverge? Which is more reliable?

3 The HML analog failed completely in our sample. Is this evidence that the value premium no longer exists, or a limitation of our methodology? How would you distinguish between these?

4 GPA has 0.93 YoY autocorrelation. What does this imply for implementation costs, and does high persistence make the signal more or less likely to be a genuine risk premium?

5 McLean & Pontiff (2016) document post-publication attenuation. Why did GPA apparently NOT attenuate in our sample? Consider survivorship, universe, and mega-cap tech dominance.

References

Financial Engineering · Lecture 04 · The Gross Profitability Premium

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